Rutherford Scattering

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The Rutherford cross section $\frac{d\sigma}{d\Omega}$ describes the probability of elastic scattering between charged particles and depends on energy and angle. The energy dependence was evaluated at the Edward Accelerator Laboratory [1] using an α (^{4}He nucleus) beam at several energies impinged on a gold foil target at forward and backward scattering angles. The backward angles exhibited excellent peak separation and allowed for the determination of the target's composition to be 80(2)% gold, 12(1)% silver, and 10(1)% copper. The extracted cross sections indicated the Rutherford cross section is proportional to $\frac{1}{E^2}$ but systematic uncertainties such as inaccurate dead time evaluations are present. Further studies will be needed to better constrain the systematic uncertainties.

I. INTRODUCTION

A fundamental observable that characterizes the interaction in nuclear physics is the cross section σ. The cross section is the probability that a scattering reaction will occur. The cross section can be viewed geometrically as the ratio between scattering events and the total of the scattered and non-scattered events:

$$\sigma = \frac{\text{Reactions per unit time}}{\text{Particles per unit time} \cdot \text{Scattering centers per unit area}}. \quad (1)$$

For short range interactions, such as hard sphere scattering, the likelihood of interacting is dependent on the interaction length. In hard sphere scattering, the cross section becomes null once the impact parameter is greater than the radius of the sphere.

However, in long-range interactions such as the Coulomb interaction whose potential is $U(r) \propto \frac{1}{r}$, the shape, strength, and range of the scattering potential is the primary factor for the interaction probability. In these long-range interactions, the cross section gains a strong energy and angle dependence of the incident particle. The total cross section can be found by integrating over all energies and angles:

$$\sigma_{\text{tot}} = \int_0^{E_{\max}} dE' \int d(\Omega) \frac{d^2\sigma(E, E', \theta)}{d\Omega dE'}. \quad (2)$$

The $\frac{d^2\sigma(E,E',\theta)}{d\Omega dE'}$ is the doubly differential cross section. In scattering experiments, one such dependence is integrated over and only one aspect of the cross section is explored, denoted by the differential cross section $\frac{d\sigma}{d\Omega}$. For brevity, the quantity of interest in this paper, the differential cross section, will be referred to as the cross section from hereafter.

II. THEORY

The description of α (^{4}He nucleus) scattering off of an atom is given by the Rutherford cross section. The Rutherford cross section is derived from the Coulomb interaction whose potential is of the form

$$U(r) = \frac{q_i q_j}{kr} \quad (3)$$

where q_k is the charge of the α particle or target atom, r is the distance between the two, and k is the proportionality constant.

The cross section in the lab frame is given by [2]

$$\frac{d\sigma}{d\Omega} = \frac{1}{4a(4\pi\epsilon_0)^2} \left(\frac{q_0 q_1}{E_0}\right)^2 \left(\frac{a + \cos\theta}{\sin^2\theta}\right)^2 \quad (4)$$

where a = $\sqrt{1 - (M_0 sin\theta / M_1)^2}$ and the index 0 (1) denote the incident (target) particle. In the limit of $M_0 \ll M_1$, the cross section can be expanded in term of M_0/M_1 to obtain

$$\frac{d\sigma}{d\Omega} = 1.296 \left(\frac{Z_0 Z_1}{E_0}\right)^2 \left(\frac{1}{\sin^4\frac{\theta}{2}} - 2\left(\frac{M_0}{M_1}\right)^2\right) \quad [\text{mb MeV}^2\,\text{sr}^{-1}]. \quad (5)$$

where the proportionality constants have been computed in terms of common nuclear physics units.

Experimentally, the cross section determines the reaction rate $R = \sigma\mathcal{L}$ where $\mathcal{L}$ is the luminosity. Therefore, the cross section can be experimentally measured by finding the ratio of incident particles to detected scattered events. However, targets used in laboratories are commonly alloys since pure targets are difficult to machine. Therefore, the target composition needs to be well understood to measure the Rutherford cross section experimentally.

The composition ratio n_A/n_B can be related to the ratio of scattering events of different elements in the alloy multiplied by the ratio in theoretical cross sections:

$$\frac{n_A}{n_B} = \frac{\text{Cts}_A}{\text{Cts}_B} \frac{d\sigma_B/d\Omega}{d\sigma_A/d\Omega}, \quad (6)$$

where A and B are different elements in the alloy target and the theoretical cross section is given by Eqn. 5.

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With the foil composition known, the cross section can then be evaluated by

$$\frac{d\sigma}{d\Omega} = \frac{N_s}{N_0 \Delta\Omega \, N_i}, \tag{7}$$

where the N_i is the number of elemental atoms in the target, N_0 is the number of incident particles, N_s is the scattered events corrected for detector dead time, and $\Delta\Omega$ is the solid angle of the detector [3].

III. EXPERIMENTAL METHODS

The Rutherford scattering experimentr was performed at the Edwards Accelerator Laboratory using a 4.5 MeV tandem Van de Graaff accelerator. An alphatross ion source was used to deliver a monoenergetic α beam onto a gold foil target. The foil is an alloy of gold, silver, and copper. The foil target was fixed at the center of an evacuated chamber. Two semiconducting silicon detectors were used to detect the scattering reaction from the foil. One detector was fixed at a backward angle of 168° beam left while the other was permitted to rotate about the target at a fixed radius. The two detectors are referred to as the "Fixed" and "Movable" detectors respectively. A schematic diagram of the experimental apparatus is provided in Fig. 1 and the foil and detector specifications are given in Table I.

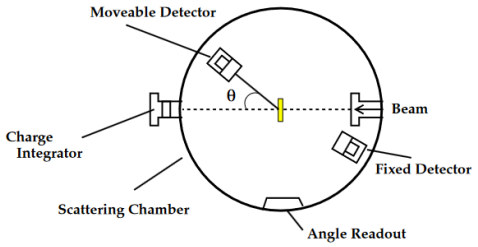


Figure 1. The RBS chamber apparatus. Adopted from [3].

To analyze the energy dependence of the Rutherford cross section, measurements were taken at several beam energies for two detector configurations, provided in Table II. The angles selected for the movable detector were done so that analysis could be conducted on forward (50°) scattering reactions and backward scattering reactions (155° and 168°).

The signals from the detector were prepared for processing by a preamplfier module. A 60 Hz pulser signal was superimposed with the detector signal allowing for voltage calibration since the pulser signal can be tuned to appear in a specific ADC bin. The detector plus pulser signal was amplified and converted to a unipolar signal by an amplifier before being readout by a 13 bit ADC. The full data acquisition system (DAQ) is shown in Fig. 2. The full run configurations are provided in Appendix A.

Type	Name	Description
Target	Composition	%
	Au	80
	Ag	10
	Cu	10
Movable	Distance	6.700 (35) cm
	Diameter	4.01 mm
	Angle Range (left)	$[-25°, -135°]$
	Angle Range (right)	$[25°, 155°]$
Fixed	Distance	6.075 (35) cm
	Diameter	3.69 mm
	Angle (fixed)	$-168°$

Table I. The nominal target and detector specifications. The gold target is an alloy of gold, silver, and copper. The detector angles are calculated clockwise relative to the direction of the beam line as shown in Fig. 1.

Type	Setting 1	Setting 2
θ_{Fixed}	168°	168°
θ_{Movable}	50°	155°
θ_{Targ}	25°	0°
Energy [MeV]	2.5, 3.5, 4.5, 6.5	

Table II. The detector configuration and the beam energies under study. The movable detector angle is measured beam right and the fixed detector is measured beam left.

IV. ANALYSIS AND RESULTS

The data collected in each run consisted of counts versus ADC bin, as shown in Fig. 3. The foil target was an alloy with majority gold but had appreciable amounts of copper and silver. For forward angle scattering, the copper and silver scattering could not be resolved from the gold scattering, resulting in a broad peak of the three scattering events superimposed. However, the backwards angle data showed clear peak separation and the elemental peaks could be resolved.

The analysis consisted of three parts. In part one, an energy calibration was constructed to identify the elemental peaks. The energy calibration was performed iteratively, starting with an ansatz for the elemental peak positions. Only spectra with distinct peak separation were used in the energy calibration analysis. The second part was a verification of the gold foil's

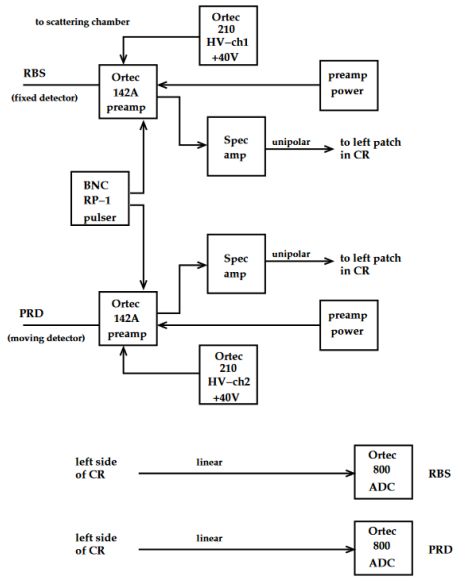


Figure 2. The data acquisition system employed for the Rutherford cross section experiment [3].

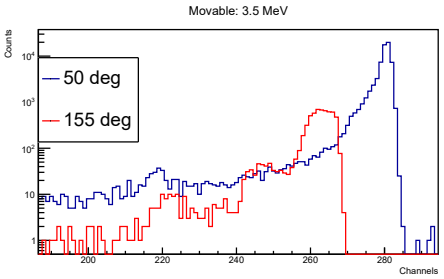


Figure 3. Counts versus ADC bin spectrum for the 3.5 MeV data set. Notice the lack of clear peak separation for the forward angle (50°) compared to the backward angle (155°).

composition. The third part was the extraction of the cross section for gold, silver, copper, or for the entire foil, depending on the separability of the peaks. Full peak information is provided in Appendix B and C.

A. Energy Calibration

An energy calibration was performed iteratively. The spectra were inspected visually and only the spectra with decent peak separation were selected. The backward angle (155°) exhibited the best peak separation across the energy range of interest (see Fig. 3). Therefore, the backward angle runs were used to obtain three peak locations that were artificially assigned the designation Cu (left-most), Ag (center), and Au (right-most) peaks. The designations were inferred from a priori knowledge of ADC binning but were confirmed by the energy calibration's low residuals.

The mean bin position and the standard deviation for each peak were calculated from the moments: $< x^n >= \frac{1}{N}\sum_i w_i x_i^n$, where N is the total counts in the peak, x_i is the ADC bin, w_i is the counts associated with the i-th bin, and n is the order of the moment. The energy calibration was constructed by plotting the mean bin position for each peak against the scattered α energy and applying a linear fit. The energy calibration was then iterated by using the constructed fit to determine the expected bin position for each element. By superimposing (see Fig. 4), the peak identification could be better constrained and the energy calibration analysis could be repeated. This was performed for the fixed and movable detector separately. The final energy calibrations and residuals for both detectors are shown in Figs 5 and 6. Both energy calibrations have residuals on the order of 10 KeV. Therefore, we can accurately identify the peaks corresponding to gold, silver, and copper. Using the energy calibration, it was confirmed that elemental peaks all fell within one broad peak for the forward angle data sets.

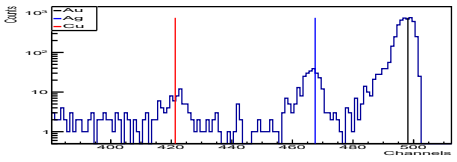


Figure 4. Peak identification using a preliminary energy calibration. The ranges of the peaks could be better constrained and the energy calibration analysis could be iterated.

B. Target Composition

With the peaks accurately determined, the run with the best peak separation was used to validate the foil composition (Table I). The 6.5 MeV run at the backward angle (155°) was determined to have the best peak separation and is shown in Fig. 3.

By imposing the condition

$$n_{\mathrm{Au}} + n_{\mathrm{Ag}} + n_{\mathrm{Cu}} = 1, \tag{8}$$

on Eqn. 6, it can be shown that

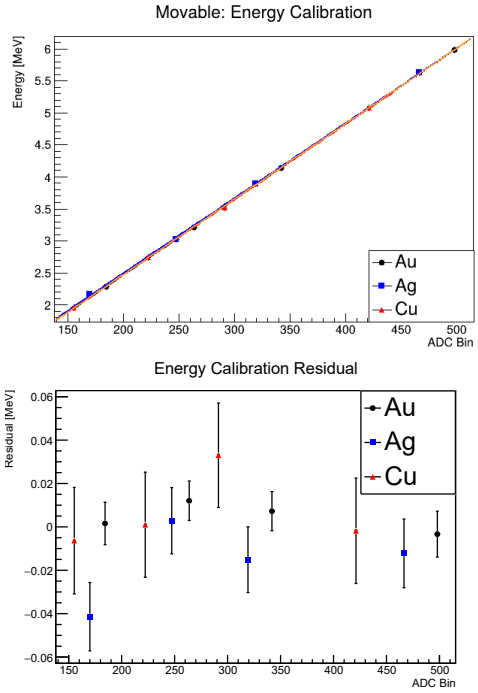


Figure 5. The final energy calibration and residuals used for the peak identification for the movable detector.

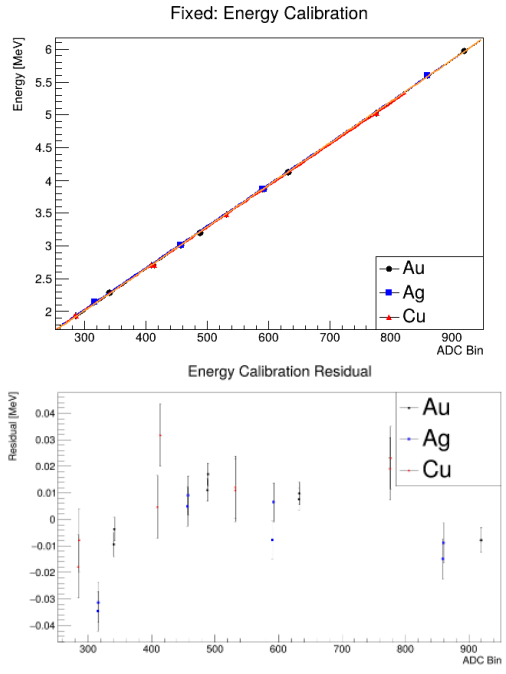


Figure 6. The final energy calibration and residuals used for the peak identification for the fixed detector

$$
\begin{aligned}
n_{\mathrm{Ag}} &= \frac{\mathrm{Cts}_{\mathrm{Ag}}}{d\sigma_{\mathrm{Ag}}/d\Omega}\left(\frac{\mathrm{Cts}_{\mathrm{Au}}}{d\sigma_{\mathrm{Au}}/d\Omega}+\frac{\mathrm{Cts}_{\mathrm{Ag}}}{d\sigma_{\mathrm{Ag}}/d\Omega}+\frac{\mathrm{Cts}_{\mathrm{Cu}}}{d\sigma_{\mathrm{Cu}}/d\Omega}\right)^{-1}\\
n_{\mathrm{Cu}} &= \frac{\mathrm{Cts}_{\mathrm{Cu}}}{d\sigma_{\mathrm{Cu}}/d\Omega}\left(\frac{\mathrm{Cts}_{\mathrm{Au}}}{d\sigma_{\mathrm{Au}}/d\Omega}+\frac{\mathrm{Cts}_{\mathrm{Ag}}}{d\sigma_{\mathrm{Ag}}/d\Omega}+\frac{\mathrm{Cts}_{\mathrm{Cu}}}{d\sigma_{\mathrm{Cu}}/d\Omega}\right)^{-1}\\
n_{\mathrm{Au}} &= \frac{\mathrm{Cts}_{\mathrm{Au}}}{d\sigma_{\mathrm{Ag}}/d\Omega}\left(\frac{\mathrm{Cts}_{\mathrm{Au}}}{d\sigma_{\mathrm{Au}}/d\Omega}+\frac{\mathrm{Cts}_{\mathrm{Ag}}}{d\sigma_{\mathrm{Ag}}/d\Omega}+\frac{\mathrm{Cts}_{\mathrm{Cu}}}{d\sigma_{\mathrm{Cu}}/d\Omega}\right)^{-1},
\end{aligned}
\tag{9}
$$

where $\frac{d\sigma}{d\Omega}$ is the theoretical cross section given by Eqn. 5. The composition of the foil was found to be in agreement with the nominal composition. The composition was found to be in excellent agreement with the target specifications. The measured compositions are given in Table III.

Element	$d\sigma/d\Omega$ [mb]	Cts	Composition
Cu	109.71 (32)	81 (9)	0.10 (1)
Ag	294.63 (52)	260 (16)	0.12 (1)
Au	845.62 (81)	4876 (69)	0.78 (2)

Table III. The composition of the gold foil. It is in good agreement with the nominal composition given in Table I.

C. Cross Section Determination

With the foil composition well understood, the Rutherford cross section can be measured. Two approaches were used depending if the peaks were separable into distinct regions or not. If peak separation could be performed, then the cross section could be found for each element within the foil. However, if the peaks were not separable, the the cross section of the entire foil was found.

1. *Separable Peak Approach*

The peaks in each backward angle data sets (155° and 168°) were separable into individual gold, silver, and copper peaks. The Rutherford cross section was calculated for each element (Au, Ag, and Cu) directly using Eqn. 7. The measured cross sections are plotted against the theoretical values in Figs. 7 and 8.

The trend in the measured cross section agree with the theoretical values. However, the measured cross sections are systematically shifted below theory. This will be discussed more in section IV C 3. To verify that the Rutherford cross section is proportional to E^2, the cross sections were multiplied by E^2 and are displayed in Figs. 9 and 10. The Rutherford cross sections become constant with respect to energy, confirming tat the cross section is proportional to $\frac{1}{E^2}$.

2. Inseparable Peak Approach

The peaks in the forward angle (50°) data sets were not separable into individual gold, silver, and copper peaks. The Rutherford cross section for the foil is then the sum of the gold, silver, and copper cross sections:

$$\frac{d\sigma_{\text{Foil}}}{d\Omega} = \frac{d\sigma_{\text{Au}}}{d\Omega} \cdot n_{\text{Au}} + \frac{d\sigma_{\text{Ag}}}{d\Omega} \cdot n_{\text{Ag}} + \frac{d\sigma_{\text{Cu}}}{d\Omega} \cdot n_{\text{Cu}}. \quad (10)$$

The $\frac{d\sigma_{\text{Foil}}}{d\Omega}$ can be calculated experimentally using Eqn. 7 and $\frac{d\sigma_{\text{Au}}}{d\Omega} \cdot n_{\text{Au}} + \frac{d\sigma_{\text{Ag}}}{d\Omega} \cdot n_{\text{Ag}} + \frac{d\sigma_{\text{Cu}}}{d\Omega} \cdot n_{\text{Cu}}$ can be calculated theoretically using Eqn. 5 and the found foil compositions listed in Table III. The measured foil cross section is plotted against the calculated theoretical cross section in Fig. 11. The measured foil cross section strongly disagree with the theoretical cross section. The poor agreement carries over to the energy dependence of the cross section. Indeed, when the measured cross section is multiplied by E^2, no discernible pattern appears. This differs greatly from the expected $\frac{1}{E^2}$ dependence. This will be discussed more in section IV C 3.

3. Sources of Systematic Error

The measured cross section for the backward angles, 155° and 168°, exhibited a $\frac{1}{E^2}$ trend as shown in Figs. 9 and 10; however, the measured cross sections were systematically lower than the predicted theoretical cross section. Alternatively, the measured cross section for the foil disagreed greatly from theory and no longer exhibited the $\frac{1}{E^2}$ trend, as shown in Fig. 11. It is highly likely that the reported dead time for the forward and backward angle measurements were inaccurate, leading to a systematic decrease in the detected counts in Rutherford scattering. This can clearly be seen in the Fig. 11, as the calculated cross section for 2.5 and 3.5 MeV appear to level off. Additionally, the 2.5 and 3.5 MeV have comparable events (see Table. VI), which is expected if the event rates exceeds the DAQ's capabilities resulting in dead time. Since dead time would impact high rate measurements, and rate is proportional to cross section, an inaccurate dead time reading would impact the forward angle cross section more than the backward angle as the expected event rate is maximal at forward angles.

However, the dead time alone cannot explain the discrepancy. Fig. 12 displays the ratio of the measured cross section to theory for the 155° backward angle. The trend in the ratios is suggestive of a energy-dependence. However, the ratio moves further away from unity at higher energy, which is counter to what is expected if dead time was the sole systematic uncertainty. This leads to the conclusion that dead time has the potential for a large systematic uncertainty at low-energy, but dead time alone cannot fully explain the systematic shift at higher energies.

V. CONCLUSION

The Rutherford cross section describes the interaction probability of a ^{4}He nucleus (α) onto a foil target. The cross section depends on angle and energy of the incident α. The energy dependence was explored using a tandem Van de Graaff accelerator at the Edwards Accelerator Laboratory by impending α's onto a gold foil target at different energies and at two detector settings. The composition of the target was analyzed and agrees well with the target specifications. The target was found to be a majority gold alloy with 78% gold, 12% silver, and 10% copper. With the foil composition known, the Rutherford cross section was found experimentally. The cross section indicated a $\frac{1}{E^2}$ energy-dependence for the backward angle measurements; however, this energy dependence could not be resolved for the forward angle measurements. The dead time has been identified as a potential source of severe systematic uncertainty that can partially explain the poor forward angle cross section extraction, but dead time alone cannot fully explain the complete systematic cross section shift observed in all data sets. Future studies are needed to better constrain the systematic uncertainties within the performed experiment.

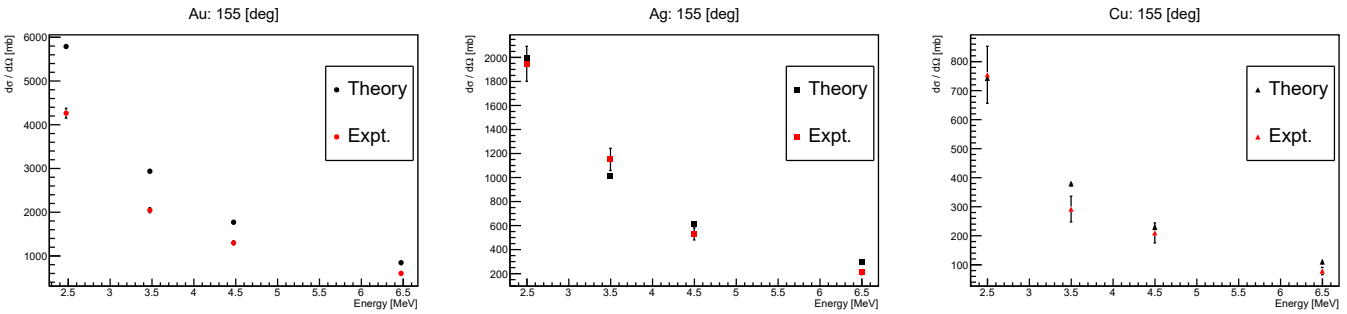


Figure 7. The extracted gold (left), silver (center) and copper (right) Rutherford cross sections for the movable detector (155°). The reported theory values are found with Eqn. 5.

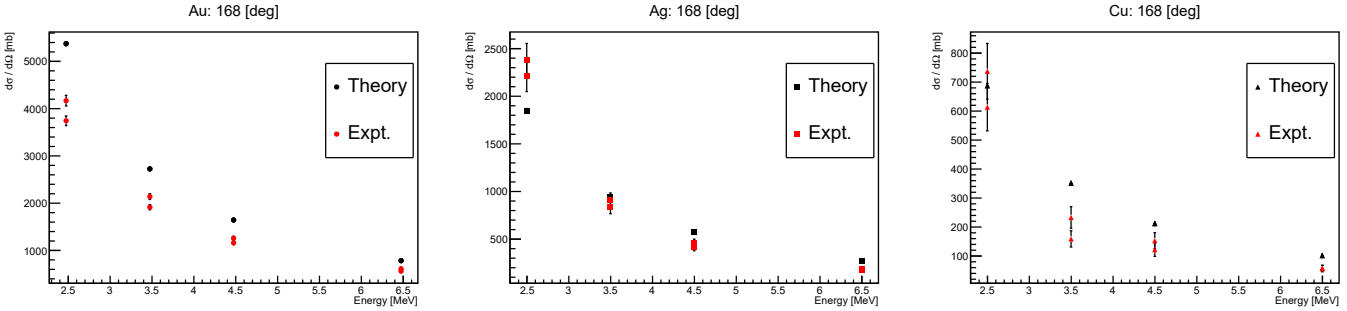


Figure 8. The extracted gold (left), silver (center) and copper (right) Rutherford cross sections for the fixed detector (168°). The doubly valued points are from the fact the fixed detector is fixed at 168° for all runs while the movable detector has two configurations for each energy. The target itself is rotated to account for the movable detector's angle. Therefore, the fixed detector detects α's that have attenuated more in the target for half of the runs. The reported theory values are found with Eqn. 5.

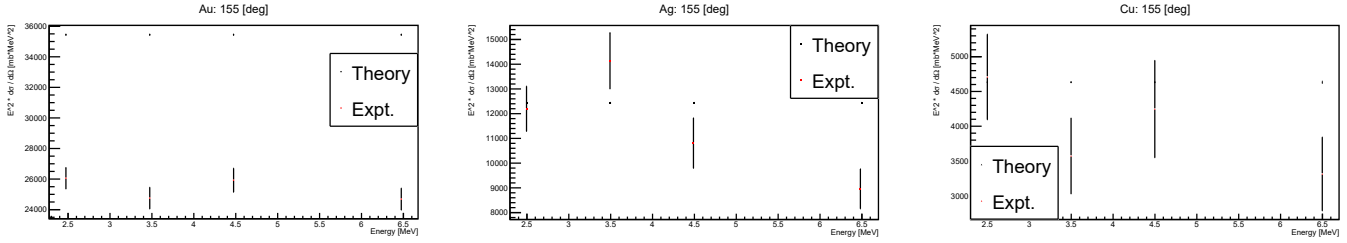


Figure 9. The extracted gold (left), silver (center) and copper (right) Rutherford cross sections multiplied by E^2 for the movable detector (155°).

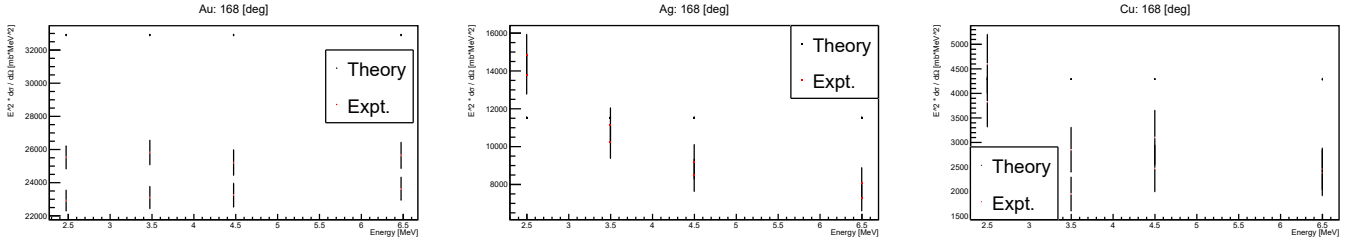


Figure 10. The extracted gold (left), silver (center) and copper (right) Rutherford cross sections multiplied by E^2 for the fixed detector (168°). The doubly valued points are from the fact the fixed detector is fixed at 168° for all runs while the movable detector has two configurations for each energy. The target itself is rotated to account for the movable detector's angle. Therefore, the fixed detector detects α's that have attenuated more in the target for half of the runs.

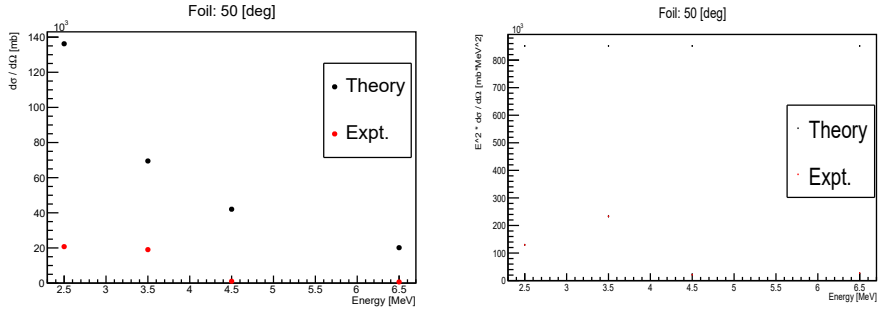


Figure 11. Left: The foil cross section calculated using Eqn. 10. Right: The foil cross section multiplied by E^2.

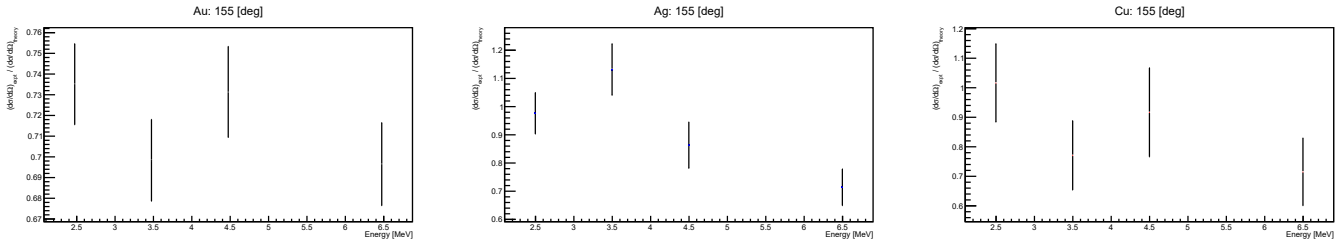


Figure 12. Ratio of the measured cross section and theory. The ratio is suggestive of an energy dependence.

[1] Ohio University *Edwards Accelerator Laboratory* (https://www.ohio.edu/cas/edwards-accelerator-laboratory) Accessed: 11/25/24

[2] D. G. Sargood, *Physics Reports 93, No. 61-116*, (1982).

[3] G.Jensen, P.M. King, and J. Roche, *PHYS6751: Nuclear Graduate Lab Lab manual v24.5* (Ohio University, 2024).

Appendix A: Run Configuration

Tag	E [MeV]	Θ_{Det}	Θ_{Targ}	I [nA]	BCI	Scale [10^{-x}]	DTL [%]	DTR [%]	Time [sec]
0020	2.5	155.0 (1)	0.0 (1)	6	5000	8	1.0465	1.1628	86
0040	2.5	50.0 (1)	25.0 (1)	6	5000	8	5.4118	1.2941	85
1010	3.5	50.0 (1)	25.0 (5)	3	5000	8	1.9417	0.72816	206
1020	3.5	155.0 (1)	0.0 (1)	3	5000	8	0.46875	0.72917	192
1030	4.5	155.0 (1)	0.0 (1)	1.5	5000	8	0.34853	0.61662	373
1040	4.5	50.0 (1)	25.0 (1)	1.3	5000	8	0.96685	0.60773	373
1050*	6.5	50.0 (1)	25.0 (1)	1.5	10000	8	1.1679	0.024331	822
1060*	6.5	155.0 (1)	0.0 (1)	1.25	15000	8	0.027027	0.027027	1110

Table IV. Run configurations. θ_{det} refers to the movable detectors as the fixed detector (not listed) is fixed at 168° beam left. DTL(R) is the dead time percentage for the movable (fixed) detector. * denotes that the pulser was turned off.

Appendix B: Peak Information - Movable Detector

Tag	Element	Bin Low	Bin High	Mean Bin	Sigma	Counts	E_{beam} [MeV]	E_{avg} [MeV]	$E_{\mathrm{scatt.}}$ [MeV]	$\theta_{\mathrm{det.}}$ [deg]
20	Cu	145	162	155.3 (3)	4.69	257 (16)	2.5	2.498	1.95 (2)	155.0 (1)
	Ag	162	1774	170.0 (1)	3.19	794 (28)	2.5	2.497	2.16 (1)	155.0 (1)
	Au	174	195	184.36 (3)	3.57	11625 (107)	2.5	2.474	2.289 (8)	155.0 (1)
40	Foil	180	200	196.04 (1)	2.95	68997 (262)	2.5	2.5	2.45 (2)	50.0 (1)
1010	Foil	260	285	280.45 (1)	2.93	65658 (256)	3.5	3.5	3.44 (2)	50.0 (1)
1020	Cu	210	230	222.3 (5)	4.73	100 (10)	3.5	3.498	2.73 (2)	155.0 (1)
	Ag	231	254	247.3 (2)	4.92	472 (21)	3.5	3.497	3.03 (1)	155.0 (1)
	Au	254	271	263.76 (4)	2.94	5634 (75)	3.5	3.474	3.21 (1)	155.0 (1)
1030	Cu	280	304	291.5 (6)	5.11	72 (8)	4.5	4.498	3.52 (2)	155.0 (1)
	Ag	305	325	319.2 (3)	4.04	218 (14)	4.5	4.497	3.90 (1)	155.0 (1)
	Au	325	350	341.9 (1)	3.39	3562 (59)	4.5	4.47	4.14 (1)	155.0 (1)
1040	Foil	345	368	363.6 (1)	3.69	3642 (60)	4.5	4.5	4.43 (2)	50.0 (1)
1050	Foil	510	532	529.1 (1)	4.22	4486 (66)	6.5	6.5	6.39 (2)	50.0 (1)
1060	Cu	408	430	421.2 (5)	4.73	81 (9)	6.5	6.497	2.705 (2)	155.0 (1)
	Ag	450	477	466.5 (3)	4.24	260 (16)	6.5	6.497	5.63 (1)	155 (1)
	Au	480	505	498.01 (4)	2.90	4876 (69)	6.5	6.474	5.99 (1)	155 (1)

Table V. Extracted peak information for the movable detector.

Appendix C: Peak Information - Fixed Detector

Tag	Element	Bin Low	Bin High	Mean Bin	Sigma	Counts	E_{beam} [MeV]	E_{avg} [MeV]	$E_{\text{scatt.}}$ [MeV]	$\theta_{\text{det.}}$ [deg]
20	Cu	272	296	285.9 (4)	6.17	215 (14)	2.5	2.498	1.93 (1)	168.0 (1)
	Ag	296	325	316.1 (2)	7.04	928 (30)	2.5	2.497	2.15 (1)	168.0 (1)
	Au	326	357	341.1 (1)	5.61	10520 (102)	2.5	2.474	2.822 (4)	168.0 (1)
40	Cu	270	296	284.3 (4)	6.63	258 (16)	2.5	2.498	1.93 (1)	168.0 (1)
	Ag	296	324	106.9 (2)	6.93	997 (31)	2.5	2.497	2.151 (7)	168.0 (1)
	Au	324	355	340.2 (1)	6.16	11700 (108)	2.5	2.474	2.822 (4)	168.0 (1)
1010	Cu	395	422	409.3 (7)	6.17	82 (90)	3.5	3.498	2.71 (1)	168.0 (1)
	Ag	442	466	456.9 (4)	5.60	383 (19)	3.5	3.497	3.01 (1)	168.0 (1)
	Au	466	501	488.1 (1)	5.77	6037 (77)	3.5	3.474	3.205 (4)	168.0 (1)
1020	Cu	405	425	413.6 (7)	5.23	56 (7)	3.5	3.498	2.71 (1)	168.0 (1)
	Ag	435	468	457.5 (4)	6.59	353 (18)	3.5	3.497	3.01 (1)	168.0 (1)
	Au	471	505	489.0 (1)	5.26	5402 (73)	3.5	3.474	3.205 (4)	168.0 (1)
1030	Cu	510	550	532.0 (13)	8.64	43 (6)	4.5	4.498	3.48 (1)	168.0 (1)
	Ag	575	604	592.3 (5)	6.07	177 (13)	4.5	4.497	3.87 (1)	168.0 (1)
	Au	604	644	632.6 (1)	5.68	3281 (57)	4.5	4.474	4.127 (4)	168.0 (1)
1040	Cu	510	550	531.8 (10)	7.62	54 (7)	4.5	4.498	3.48 (1)	168.0 (1)
	Ag	560	604	590.0 (5)	7.54	192 (13)	4.5	4497	3.87 (1)	168.0 (1)
	Au	604	645	632.3 (1)	5.62	3559 (59)	4.5	4.474	4.124 (4)	168.0 (1)
1050	Cu	756	800	776.7 (1.6)	10.02	40 (6)	6.5	6.497	5.60 (1)	168.0 (1)
	Ag	830	875	859.1 (5)	6.37	162 (12)	6.5	6.497	5.60 (1)	168.0 (1)
	Au	892	932	919.3 (1)	5.00	3478 (58)	6.5	6.474	5.972 (4)	168.0 (1)
1060	Cu	755	790	776.1 (1.1)	8.66	62 (7)	6.5	6.498	5.60 (1)	168.0 (1)
	Ag	845	870	860.0 (3)	4.58	220 (14)	6.5	6.497	5.60 (1)	168.0 (1)
	Au	870	930	919.3 (1)	5.99	4808 (69)	6.5	6.474	5.972 (4)	168.0 (1)

Table VI. Extracted peak information for the fixed detector.